

It's



Easy Dude.

↳ I forget to find
eigenspace in some QW,
but I've found them in
a few so you can take
reference from those.

V-4Eigen value & Eigen vectorBefore
CIE-2

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1) A^{-1} if $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$.

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Characteristic eqn is given by,

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0 \quad \text{--- I}$$

where,

$$S_1 = \text{Tr}(A) = 1 + 2 - 1 = 2$$

$$\begin{aligned} S_2 &= \text{Sum of Cofactor of diagonal elements of } A \\ &= -2 - 1 - 1 + 0 + 2 + 1 \\ &= -1 \end{aligned}$$

$$\begin{aligned} S_3 &= \text{Det}(A) \\ &= -2 \end{aligned}$$

Thus,

eqn I becomes,

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0 \quad \text{--- II}$$

Now by Cayley Hamilton Theorem, every square matrix satisfy it's own characteristic eqn
i.e.

$$A^3 - 2A^2 - A + 2I = 0$$

divide by A

$$A^2 - 2A - I + 2A^{-1} = 0$$

$$\text{or, } A^{-1} = \frac{1}{2} (-A^2 + 2A + I)$$

$$\text{or, } A^{-1} = \frac{1}{2} \left(\begin{bmatrix} 0 & -1 & -1 \\ +3 & -4 & -3 \\ +1 & -1 & -2 \end{bmatrix} + \begin{bmatrix} 2 & 2 & -4 \\ -2 & 4 & 2 \\ 0 & 2 & -2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

$$\therefore A^{-1} = \begin{bmatrix} 3/2 & 1/2 & -5/2 \\ 1/2 & 1/2 & -1/2 \\ 1/2 & 1/2 & -3/2 \end{bmatrix}$$

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2) A^4 given $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$.

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↳ Characteristic eqⁿ

$$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0 \quad \text{--- I}$$

So,

$$s_1 = 11$$

$$s_2 = -1 - 3 + 0 = -4$$

$$s_3 = -1$$

from eqⁿ I

$$\lambda^3 - 11\lambda^2 - 4\lambda + 1 = 0$$

Now by Cayley Hamilton Theorem every square matrix satisfy its own C.E.
i.e.

$$A^3 - 11A^2 - 4A + I = 0$$

Multiplying by A

$$A^4 - 11A^3 - 4A^2 + A = 0$$

$$\text{or, } A^4 = 11A^3 + 4A^2 - A$$

$$\begin{aligned} \text{or, } A^4 &= 11 \begin{bmatrix} 157 & 283 & 353 \\ 283 & 510 & 636 \\ 353 & 636 & 793 \end{bmatrix} + 4 \begin{bmatrix} 14 & 25 & 31 \\ 25 & 45 & 56 \\ 31 & 56 & 70 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 1782 & 3211 & 4004 \\ 3211 & 5789 & 7215 \\ 4004 & 7215 & 8997 \end{bmatrix} \end{aligned}$$

3) A^{-1} given $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$. Also Compute A^8 .

↳ C.E. = $\lambda^2 - s_1 \lambda + s_2 = 0 \quad \text{--- I}$

So $s_1 = \text{Tr}(A) = 0$

$$s_2 = |A| = -5$$

from eqⁿ I

$$\lambda^2 - 5 = 0$$

By Cayley Hamilton Theorem,
 $A^2 - 5I = 0 \quad \text{--- II}$

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Multiplying by A^{-1}

$$A - 5A^{-1} = 0$$

$$\text{or, } A^{-1} = \frac{1}{5} A$$

$$\therefore A^{-1} = \begin{bmatrix} 1/5 & 2/5 \\ 2/5 & -1/5 \end{bmatrix} //$$

Similarly for A^4 from eqn I

$$A^4 = A^2 \cdot A^2$$

$$= 5I \cdot 5I$$

$$= \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}$$

So,

$$A^8 = A^4 \cdot A^4$$

$$= \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix} \cdot \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}$$

$$\therefore A^8 = \begin{bmatrix} 625 & 0 \\ 0 & 625 \end{bmatrix} //$$

4) Find A^{-1} , A^{-2} , A^{-3} if $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$.

→ Given, $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$

C.E. $\Rightarrow \lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0 \quad \text{--- I}$

So,

$$s_1 = 4,$$

$$s_2 = 1 + 2 + 2 = 5$$

$$s_3 = 4$$

So, from I

$$\lambda^3 - 4\lambda^2 + 5\lambda - 4 = 0 \quad \text{--- II}$$

By Cayley Hamilton Theorem

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$$A^3 - 4A^2 + 5A - 4I = 0$$

multiply by A^{-1}

$$A^2 - 4A + 5I - 4A^{-1} = 0$$

$$\therefore, A^{-1} = \frac{1}{4} [A^2 - 4A + 5I]$$

$$\therefore, A^{-1} = \frac{1}{4} \left[\begin{bmatrix} 4 & 3 & 2 \\ 2 & 1 & 4 \\ 3 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 8 & 4 & 0 \\ 0 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \right]$$

$$\therefore A^{-1} = \frac{1}{4} \begin{bmatrix} -1 & -1 & 2 \\ 2 & 2 & -4 \\ -1 & 1 & 2 \end{bmatrix}$$

Similarly,

$$A^{-2} = \frac{1}{16} \begin{bmatrix} -3 & -1 & 10 \\ 10 & -2 & -12 \\ -1 & 5 & -2 \end{bmatrix}$$

Also,

$$\begin{aligned} A^{-3} &= A^{-1} \cdot A^{-2} \\ &= \frac{1}{64} \begin{bmatrix} -15 & 11 & 18 \\ 18 & -26 & 4 \\ 11 & 9 & -26 \end{bmatrix} \end{aligned}$$

5) Given $M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ prove that $(4M - 3I)^{-2} = M^{-4}$.

$$\hookrightarrow M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, (4M - 3I)^{-2} = M^{-4}$$

$$C.E \Rightarrow A^2 - \text{tr}(M)A + |M|I = 0$$

$$\Rightarrow A^2 - 4A + 3I = 0$$

By Cayley-Hamilton theorem —

$$M^2 - 4M + 3I = 0$$

$$\therefore, M^2 = 4M - 3I \quad \text{--- } I$$

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$$\begin{aligned} \text{LHS} &= (AM - 3I)^{-2} \\ &= (M^2)^{-2} \\ &= M^{-4} = \text{RHS} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

Proved

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6) If $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ then Evaluate the expression $A^8 - A^7 + A^2 + I$.

→ Given, $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

$$\text{C.E.} \Rightarrow \lambda^2 - \text{Tr}(A) \cdot \lambda + |A| = 0$$

$$\text{or, } \lambda^2 - 1 = 0 \quad \text{--- I}$$

By Cayley -

$$A^2 - I = 0$$

$$\therefore A^2 = I$$

Now,

$$A^8 - A^7 + A^2 + I = (A^2)^4 - (A^2)^3 \cdot A + A^2 + I$$

$$= I - AI + I + I$$

$$= 3I - A$$

$$= \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -2 \\ 0 & 4 \end{bmatrix}$$

7) Given $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ find M^{-2} .

→ C.E., $\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0$

$$\text{or, } \lambda^3 - 4\lambda^2 + 4\lambda - 1 = 0$$

By Cayley.

$$M^3 - 4M^2 + 4M - I = 0 \quad \text{--- I}$$

multiplying by M^{-1}

$$M^2 - 4M + 4I - M^{-1} = 0$$

$$\therefore M^{-1} = M^2 - 4M + 4I \quad \text{--- } \square$$

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Multiplying by M^{-1}

$$M^{-2} = M - 4I + 4M^{-1} \quad \text{--- II'}$$

Substitute M^{-1} from eqn I to III

$$M^{-2} = M - 4I + 4(M^2 - 4M + 4I)$$

$$\text{or, } M^{-2} = M - 4I + 4M^2 - 16M + 16I$$

$$\text{or, } M^{-2} = 4M^2 - 15M + 12I$$

$$\text{or, } M^{-2} = 4 \left[\begin{array}{ccc|ccc} 2 & 3 & 0 & 15 & 15 & 0 \\ 3 & 5 & 0 & 15 & 30 & 0 \\ 0 & 0 & 1 & 0 & 0 & 15 \end{array} \right] + \left[\begin{array}{ccc|ccc} 12 & 0 & 0 & 0 & 12 & 0 \\ 0 & 12 & 0 & 0 & 0 & 12 \end{array} \right]$$

$$\therefore M^{-2} = \left[\begin{array}{ccc|ccc} 5 & -3 & 0 & 15 & 15 & 0 \\ -3 & 2 & 0 & 15 & 30 & 0 \\ 0 & 0 & 1 & 0 & 0 & 15 \end{array} \right] //$$

8) Solve the following system of linear equations $AX=B$ by finding the inverse of the coefficient

$$2x - y + z = 2$$

$$\text{matrix using the Cayley-Hamilton Theorem: } 3x + y - 2z = 9$$

$$-x + 2y + 5z = -5$$

$$\rightarrow L.E \Rightarrow AX=B \quad \text{--- I}$$

$$\text{So, } A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & 1 & -2 \\ -1 & 2 & 5 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 2 \\ 9 \\ -5 \end{bmatrix}$$

now,

$$C.E, \lambda^3 - s_1\lambda^2 + s_2\lambda - s_3 = 0 \quad \text{--- II}$$

$$s_1 = 8$$

$$s_2 = 9 + 11 + 5 = 25$$

$$s_3 = 38$$

from eqn I,

$$\lambda^3 - 8\lambda^2 + 25\lambda - 38 = 0$$

By Cayley,

$$A^3 - 8A^2 + 25A - 38I = 0 \quad \text{--- III}$$

Multiplying by A^{-1}

$$A^2 - 8A + 25I - 38A^{-1} = 0$$

$$\text{or, } A^{-1} = \frac{1}{38} [A^2 - 8A + 25I]$$

$$\text{or, } A^{-1} = \frac{1}{38} \left[\begin{bmatrix} 0 & -1 & 9 \\ 11 & -6 & -9 \\ -1 & 13 & 20 \end{bmatrix} - \begin{bmatrix} 16 & -8 & 8 \\ 24 & 8 & -16 \\ -8 & 16 & 40 \end{bmatrix} + \begin{bmatrix} 25 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 25 \end{bmatrix} \right]$$

$$\therefore A^{-1} = \frac{1}{38} \begin{bmatrix} 9 & 7 & 1 \\ -13 & 11 & 7 \\ 7 & -3 & 5 \end{bmatrix}$$

now, from eqn I

$$AX = B$$

$$\text{or, } X = A^{-1}B$$

$$\text{or, } X = \frac{1}{38} \begin{bmatrix} 9 & 7 & 1 \\ -13 & 11 & 7 \\ 7 & -3 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 9 \\ -5 \end{bmatrix}$$

$$\text{or, } X = \frac{1}{38} \begin{bmatrix} 76 \\ 38 \\ -38 \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} //$$

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9) If $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 4 & 0 & 1 \\ 3 & 1 & 5 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ and $\lambda_1 = 1$ and $\lambda_2 = 2$ then is the matrix invertible?

ANS: $\lambda_3 = 4, \lambda_4 = 5$.

WKT, $\text{Tr}(A) = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4$

$$\text{or, } 12 = 1 + 2 + \lambda_3 + \lambda_4$$

$$\therefore \lambda_3 + \lambda_4 = 9 \quad \text{--- I}$$

Also

$$|A| = \lambda_1 \cdot \lambda_2 \cdot \lambda_3 \cdot \lambda_4$$

$$\text{or, } 2 \times 20 \times \begin{vmatrix} 4 & 0 & 1 \\ 1 & 5 & 2 \\ 0 & 0 & 1 \end{vmatrix} = 2 \lambda_3 \cdot \lambda_4$$

$$\text{or, } 2 \times 20 = 2 \lambda_3 \lambda_4$$

$$\therefore \lambda_3 \lambda_4 = 20 \quad \text{--- I}$$

$$\begin{aligned} \text{Thus, } (\lambda_3 - \lambda_4)^2 &= (\lambda_3 + \lambda_4)^2 - 4\lambda_3\lambda_4 \\ &= 81 - 4 \times 20 \\ &= 1 \end{aligned}$$

$$\therefore \lambda_3 - \lambda_4 = 1 \quad \text{--- II}$$

from eqn I & II

$$\lambda_3 = 5$$

$$\lambda_4 = 4$$

Since

$\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are non-zero

$\therefore$ invertible

10) If $A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 1 & 2 & 0 & 5 \\ 0 & 4 & 2 & 0 \\ 2 & 4 & 0 & 0 \end{bmatrix}$ and $\lambda_1 = -4$ and $\lambda_2 = 2$ then is the matrix invertible?

ANS: $\lambda_3 = 0, \lambda_4 = 7$.

$$\hookrightarrow \text{Tr}(A) = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4$$

$$\text{or, } \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 5$$

$$\therefore \lambda_3 + \lambda_4 = 7 \quad \text{--- I}$$

Also,

$$\det(A) = \lambda_1 \cdot \lambda_2 \cdot \lambda_3 \cdot \lambda_4$$

$$\text{or, } 2 \cdot \det \begin{vmatrix} 1 & 2 & 4 \\ 1 & 2 & 5 \\ 2 & 4 & 0 \end{vmatrix} = -8 \lambda_3 \lambda_4$$

$$\text{or, } \lambda_3 \lambda_4 = 0$$

$$\therefore \lambda_3 = 0$$

from I

$$\lambda_4 = 7$$

Since,

$\lambda_3 = 0$ (one of the eigen value be 0)

$\therefore$ not invertible

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IV. Eigen Spaces:

Theorem: (The eigenspace of A corresponding to λ)

If $T: V \rightarrow V$ is a linear transformation then the subset of all eigenvectors corresponding to λ along with the zero vector is a subspace of V and is called the eigen space of λ denoted by $E(\lambda)$.

Note:

1. If an eigenvalue λ occurs as a multiple root (k times) of the characteristic polynomial, then λ has **algebraic multiplicity** (A.M.) k .

Eigenvectors

2. The number of independent eigenvectors corresponding to a given eigenvalue is the geometric multiplicity (G.M.) and hence the dimension of the eigenspace of an eigenvalue λ of A is the **geometric multiplicity** of λ .
3. $0 < G.M. \leq A.M.$ and if $A.M. = 1$ then $G.M. = 1$.
4. The eigenvalue λ is said to be defective if the geometric multiplicity of λ is less than its algebraic multiplicity.
5. An $n \times n$ matrix A is said to be defective if it doesn't have n linearly independent eigenvectors. Equivalently, an $n \times n$ matrix A is said to be defective if it has a defective eigenvalue.

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Obtain algebraic and geometric multiplicity of the eigenvalues for the following linear transformations (or Matrices) by finding the eigenspaces and hence identify the defective eigenvalues/matrices.

1) $A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$.

$$\hookrightarrow C.E. \quad \lambda^2 - \text{Tr}(A)\lambda + |A| = 0$$

$$\therefore \lambda^2 - 2\lambda + 3 = 0$$

$$\therefore \lambda_1 = 2 + \sqrt{2}i$$

$$\therefore \lambda_2 = 2 - \sqrt{2}i$$

So,

for λ_1 ,

$$A.M. = 1$$

$$G.M. = 1$$

for λ_2 ,

$$A.M. = 1$$

$$G.M. = 1$$

Since,

$A.M. = G.M.$ for every Eigenvalue.

$\therefore$ no defective eigenvalues. So matrix A is non-defective

now, subspace for $d = 1 + \sqrt{2}i$

$$A - dI = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 + \sqrt{2}i & 0 \\ 0 & 1 + \sqrt{2}i \end{bmatrix}$$

$$= \begin{bmatrix} -\sqrt{2}i & -1 \\ 2 & -\sqrt{2}i \end{bmatrix}$$

$$R_2 \rightarrow R_2 - i\sqrt{2} R_1$$

$$\approx \begin{bmatrix} -\sqrt{2}i & -1 \\ 0 & 0 \end{bmatrix}$$

no. of free variables $= 2 - 1 = 1$

Let,

$$y = k$$

then,

$$-\sqrt{2}i x - y = 0$$

$$\therefore x = -\frac{k}{\sqrt{2}i}$$

$$\therefore x = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -k/\sqrt{2}i \\ k \end{bmatrix} = \frac{-k}{\sqrt{2}i} \begin{bmatrix} 1 \\ -i\sqrt{2} \end{bmatrix}$$

$$\therefore \text{Basis} = \begin{bmatrix} 1 \\ -i\sqrt{2} \end{bmatrix} //$$

Similarly subspace for $d = 1 - i\sqrt{2}$

$$\therefore \text{Basis} = \begin{bmatrix} 1 \\ i\sqrt{2} \end{bmatrix}$$

$$2) A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

↳ C.E.

$$\lambda^2 - \text{tr}(A)\lambda + |A| = 0$$

$$\text{or, } \lambda^2 - 2\lambda + 2 = 0$$

$$\therefore \lambda_1 = 1 + i$$

$$\therefore \lambda_2 = 1 - i$$

Since, $AM = GM = 1$ for every eigenvalue.

$\therefore$ Matrix A is non-defective.

Basis for $\lambda = 1 + i$

$$A - \lambda_1 I = \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix}$$

$$R_2 \rightarrow R_2 - i R_1$$

$$\approx \begin{bmatrix} -i & -1 \\ 0 & 0 \end{bmatrix}$$

$$\therefore \text{Basis} = \begin{bmatrix} 1 \\ -i \end{bmatrix} \text{ \& } \begin{bmatrix} 1 \\ i \end{bmatrix} //$$

$$3) T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ defined by } T(x, y) = (3x + 3y, x + 5y).$$

↳ Standard basis vector of $\mathbb{R}^2 = (1, 0), (0, 1)$

Thus,

$$T(1, 0) = (3, 1)$$

$$T(0, 1) = (3, 5)$$

So,

$$A = \begin{bmatrix} 3 & 3 \\ 1 & 5 \end{bmatrix}$$

$$\text{C.E.} \Rightarrow \lambda^2 - \text{tr}(A)\lambda + |A| = 0$$

$$\text{or, } \lambda^2 - 8\lambda + 12 = 0$$

$$\therefore \lambda_1 = 6$$

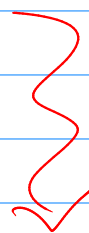
$$\therefore \lambda_2 = 2$$

Since, $AM = GM = 1$ for each eigenvalue

$\therefore$ Matrix A is non-defective.

find Basis

for all



$$4) T: \mathbb{R}^3 \rightarrow \mathbb{R}^3 \text{ defined by } T(x, y, z) = (x - y + z, x - y + z, x - 2y + 2z)$$

↳ Standard Basis vector of $\mathbb{R}^3$

$$(1, 0, 0)$$

$$(0, 1, 0)$$

$$(0, 0, 1)$$

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Thus,

$$T(1, 0, 0) = (1, 1, 1)$$

$$T(0, 1, 0) = (-1, -1, -2)$$

$$T(0, 0, 1) = (1, 1, 2)$$

$$S_0, \quad A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -1 & 1 \\ 1 & -2 & 2 \end{bmatrix}$$

$$C.E. \Rightarrow \lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0 \quad \text{--- I}$$

$$S_1 = 2$$

$$S_2 = 0 + 1 + 0 = 1$$

$$S_3 = 0$$

eqn I become,

$$\lambda^3 - 2\lambda^2 + \lambda = 0$$

$$\therefore \lambda = 0, 1, 1$$

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for $\lambda_1 = 0$	for $\lambda_2 = 1$
$A \cdot M = 1$	$A \cdot M = 2$
$G \cdot M = 1$	$G \cdot M =$

$$A - \lambda I = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -1 & 1 \\ 1 & -2 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 1 \\ 1 & -2 & 1 \\ 1 & -2 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 0 & -1 & 1 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Note:

$G \cdot M = \text{no. of free variable}$

Since

$$\text{Rank} = 2 = n$$

$$\text{no. of unknown} = r = 3$$

$$\therefore \text{no. of free variable} = 3 - 2 = 1$$

$$\therefore G \cdot M = 1.$$

Since $G \cdot M < A \cdot M$ for $\lambda = 1$.

$\therefore$ it is a defective eigenvalue.

$\therefore$ Matrix is defective

$$5) A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\rightarrow C.E. \Rightarrow \lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

where,

$$S_1 = 2$$

$$S_2 = 0 + 1 + 0 + 0 + 1 - 2 = 0$$

$$S_3 = 0$$

Thus,

$$\lambda^3 - 2\lambda^2 = 0$$

$$\therefore \lambda = 0, 0, 2$$

So,

for $\lambda_1 = 0$	for $\lambda_2 = 2$
A.M = 2	A.M = 1
G.M = 0	G.M = 1

Since,

$$G.M < A.M \text{ for } \lambda = 0$$

$\therefore$ The matrix is defective

$$6) A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

$$\rightarrow C.E. \Rightarrow \lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

So,

$$S_1 = 5$$

$$S_2 = 2 + 4 - 1 + 2 = 7$$

$$S_3 = 3$$

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Thus,

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

$$\therefore \lambda = 3, 1, 1$$

Now,

for $\lambda_1 = 3$	for $\lambda_2 = 1$
A.M = 1	A.M = 2
G.M = 1	

Now, G.O.M for λ_2

$$A - \lambda_2 I = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$\approx \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{no. of free variables} = 3 - 1 = 2 = \text{G.O.M}$$

Since A.O.M = G.O.M for all eigenvalues.
 $\therefore$ non-defective.

7) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (2x - 5y, 2y, -x + 4y + z)$.

→ Standard Basis of $\mathbb{R}^3$

$$(1 \ 0 \ 0)$$

$$(0 \ 1 \ 0)$$

$$(0 \ 0 \ 1)$$

→ Then,

$$T(1 \ 0 \ 0) = (2 \ 0 \ -1)$$

$$T(0 \ 1 \ 0) = (-5 \ 2 \ 4)$$

$$T(0 \ 0 \ 1) = (0 \ 0 \ 1)$$

So,

$$A = \begin{bmatrix} 2 & -5 & 0 \\ 0 & 2 & 0 \\ -1 & 4 & 1 \end{bmatrix}$$

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$C \in$,

$$S_1 = 5$$

$$S_2 = 2 + 2 + 4 = 8$$

$$S_3 = 4$$

Now,

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\therefore \lambda = 1, 2, 2$$

for $\lambda_1 = 1$	for $\lambda_2 = 2$
$A \cdot M = 2$	$A \cdot M = 2$
$G \cdot M = 1$	$G \cdot M = 2$

$$A - \lambda_2 I = \begin{bmatrix} 2 & -5 & 0 \\ 0 & 2 & 0 \\ -1 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -5 & 0 \\ 0 & 0 & 0 \\ -1 & 4 & -1 \end{bmatrix}$$

$$\text{free variable} = 3 - 2 = 1$$

$$\therefore G \cdot M = 1$$

Since $G \cdot M < A \cdot M$ for $\lambda_2 = 2$

$\therefore$ defective matrix

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8) $T: P_1 \rightarrow P_1$ defined by $T(at + b) = (3a + 5b)t - (2a + 4b)$.

~~Standard Basis,~~

$$(a, b) = (1, 0)$$

$$(a, b) = (0, 1)$$

Thus,

$$\text{when } (a, b) = (1, 0)$$

$$T(t) = 3t - 2$$

$$\text{when } (a, b) = (0, 1)$$

$$T(1) = 5t - 4$$

So,

$$A = \begin{bmatrix} 3 & 5 \\ -2 & -4 \end{bmatrix}$$

$$C.E. \quad \lambda^2 + \lambda - 2 = 0$$

$$\therefore \lambda = 1, -2$$

for $\lambda_1 = 1$	for $\lambda_2 = -2$
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$$A \cdot M = 1$$

$$A \cdot M = 1$$

$$G \cdot M = 1$$

$$G \cdot M = 1$$

$\therefore$ no defective eigenvalue & matrix.

9) $T: P_2 \rightarrow P_2$ defined by $T(at^2 + bt + c) = (2a - c)t^2 + (2a + b - 2c)t + (-a + 2c)$.

→ when $a = 1, b = 0, c = 0$

$$T(t^2) = 2t^2 + 2t - 1$$

when $a = 0, b = 1, c = 0$

$$T(t) = 0t^2 + t + 0$$

when $a = 0, b = 0, c = 1$

$$T(1) = -t^2 - 2t + 2$$

Thus,

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 2 & 1 & -2 \\ -1 & 0 & 2 \end{bmatrix}$$

C.E.

$$\lambda^3 - 5\lambda^2 + 5\lambda - 3 = 0$$

$$\text{So, } \lambda_1 = 3$$

$$\lambda_2 = 2 + 3 + 2 = 7$$

$$\lambda_3 = 3$$

Thus,

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

$$\therefore \lambda = 3, 1, 1$$

for $\lambda = 3$ for $\lambda = 1$

$$A - \lambda I = 2$$

$$A - \lambda I = 2$$

$$G.O.M = 1$$

$$G.O.M = 2$$

$$A - \lambda I = \begin{bmatrix} 2 & 0 & -1 \\ 2 & 1 & -2 \\ -1 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ -1 & 0 & 1 \end{bmatrix}$$

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$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 + R_1$$

$$\approx \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \text{G.M. for } (\lambda = 1) = 2$$

Since, A.M. = G.M. for every eigenvalue

$\therefore$ Matrix is non-defective.

$$10) T: R^3 \rightarrow R^3 \text{ defined by } T = \begin{bmatrix} 2 & 0 & 0 \\ -1 & -1 & 9 \\ 0 & -1 & 5 \end{bmatrix}$$

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$\rightarrow$ C.E. is,

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

where,

$$S_1 = 6$$

$$S_2 = 4 + 10 - 2 = 12$$

$$S_3 = 8$$

So,

$$\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$$

$$\therefore \lambda = 2, 2, 2$$

for $\lambda = 2$

$$\text{A.M.} = 3$$

now,

G.M. for $\lambda = 2$

$$A - \lambda I = \begin{bmatrix} 0 & 0 & 0 \\ -1 & -3 & 9 \\ 0 & -1 & 3 \end{bmatrix}$$

$$\therefore \text{G.O.M.} = 3 - 2 = 1$$

Since

A.M. $\neq$ G.O.M. for all $\lambda = 2$

$\therefore$ Matrix is defective

$$11) A = \begin{bmatrix} -1 & 1 & 1 \\ -1 & 1 & 1 \\ -2 & 2 & 1 \end{bmatrix}$$

→ C.E. is,

$$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0$$

where,

$$s_1 = 1$$

$$s_2 = -1 + 1 + 0 = 0$$

$$s_3 = 0$$

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True

$$\lambda^3 - \lambda^2 = 0$$

$$\therefore \lambda = 1, 0, 0$$

for $\lambda = 1$

$$A.M = 1$$

$$G.M = 1$$

for $\lambda = 0$

$$A.M = 2$$

$$G.M = 2$$

$$A - \lambda I = \begin{bmatrix} -1 & 1 & 1 \\ -1 & 1 & 1 \\ -2 & 2 & 1 \end{bmatrix}$$

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$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\approx \begin{bmatrix} -1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$G.M = n - r = 3 - 2 = 1$$

Since, $A.M \neq G.M$ for $\lambda = 0$

$\therefore$ Matrix A is defective

- 12) Given $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and eigenvalues of M are $\lambda = 0, 2$. Find the algebraic and geometric multiplicity of each λ and hence find eigenspace corresponding to each eigenvalue of $A = M^2 + \frac{1}{2}M$.

Given,

$$M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

C.E.s

$$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0$$

where,

$$s_1 = 4$$

$$s_2 = 2 + 2 + 0 = 4$$

$$s_3 = 0$$

Thus,

$$\lambda^3 - 4\lambda^2 + 4\lambda = 0$$

$$\therefore \lambda(\lambda^2 - 4\lambda + 4) = 0$$

$$\therefore \lambda(\lambda - 2)^2 = 0$$

$$\therefore \lambda = 0, 2, 2$$

So,

for $\lambda = 0$

$$A.M = 1$$

$$G.M = 1$$

for $\lambda = 2$

$$A.M = 2$$

$$G.M = ?$$

$$M - \lambda I = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_1$$

$$\approx \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r = 1$$

$$n = 3$$

$$G.M = n - r = 2 \text{ free variable}$$

$$\text{Let, } z = k_1$$

$$y = k_2$$

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$$\text{So, } -x + y = 0$$

$$\therefore x = k_2$$

Thus,

$$x_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} k_2 \\ k_2 \\ k_1 \end{bmatrix} = k_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \text{ or, } k_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\therefore \text{eigen vector} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Similarly, eigen vectors for $\lambda = 0$

$$M - \lambda I = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$\approx \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$n - r = 3 - 2 = 1 \text{ free variable}$$

$$\text{Let, } y = k$$

So,

$$x + y = 0$$

$$\therefore x = -k$$

$$\therefore z = 0$$

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So,

$$x_2 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -k \\ k \\ 0 \end{bmatrix} = k \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\therefore \text{eigen vector (for } \lambda = 0) = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Thus,

$$\text{for } \lambda = 0 \quad \text{for } \lambda = 2$$

$$A \cdot M = 1$$

$$A \cdot M = 2$$

$$Q \cdot M = 1$$

$$Q \cdot M = 2$$

$\therefore$ Matrix M is non-defective

Now,

$$A = M^2 + \frac{1}{2} M.$$

w.k.t, λ is eigenvalue of M .
Thus,

$$f(\lambda) = \lambda^2 + \frac{1}{2} \lambda$$

for $\lambda = 0$,

$$\lambda A = 0$$

for $\lambda = 2$

$$\lambda A = 4 + 1 = 5$$

So,

eigenvalues of A are 0 & 5

Since

A is the polynomial of matrix M .

So, even though eigen values get change, but
it does not change eigen vectors.

Thus,

Eigen space of A corresponding to $\lambda A = 0$

$$= \text{Span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Eigen space of A corresponding to $\lambda A = 5$

$$= \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

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- 13) Let $f(t) = t^3 + 2t^2 + 3$ be the polynomial of degree 3 and X is the eigen vector of A corresponding to eigen value λ . Prove that X is the eigenvector of $f(A)$ corresponding to eigenvalue $f(\lambda)$.

To prove: X is eigenvector of $f(A)$ corresponding to eigenvalue $f(\lambda)$.
i.e.

$$f(A)X = f(\lambda)X$$

→ Given,

A Matrix 'A' has eigen vector = X and eigen value = λ

$$f(\lambda) = \lambda^3 + 2\lambda^2 + 3$$

Thus,

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$$f(A) = A^3 + 2A^2 + 3I$$

Multiply the matrix $f(A)$ by the vector X .

$$f(A) \cdot X = A^3X + 2A^2X + 3IX \quad \text{--- I}$$

$$\text{w.k.t. } AX = \lambda X \quad \text{--- II}$$

Thus from eqn I & II

$$\begin{aligned} A^3X &= A^2(AX) & A^2\lambda &= A \cdot \lambda X \\ &= A^2(\lambda X) & &= \lambda^2 X \\ &= A\lambda(A X) \\ &= A\lambda \cdot \lambda X \\ &= \lambda^2(A X) \\ &= \lambda^3 X \end{aligned}$$

Eqn I become

$$f(A) \cdot X = \lambda^3 X + 2\lambda^2 X + 3X$$

$$\therefore f(A) \cdot X = X(\lambda^3 + 2\lambda^2 + 3)$$

$$\therefore f(A) \cdot X = X \cdot f(\lambda)$$

//

14) $D: V \rightarrow V$ defined by $D(f) = \frac{df}{dt}$, where V is the space of functions with basis $S = \{\sin t, \cos t\}$.

To find matrix and its eigen values.

→ Given.

Basis, $S = \{\sin t, \cos t\}$

Now,

for first column

$$\begin{aligned} D(\sin t) &= \frac{d(\sin t)}{dt} \\ &= \cos t \end{aligned}$$

In terms of basis,

$$\cos t = 0(\sin t) + 1(\cos t)$$

Thus,

$$\text{First column} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Similarly for second column,

$$\begin{aligned} D(\cos t) &= \frac{d(\cos t)}{dt} \\ &= -\sin t \end{aligned}$$

In terms of basis,

$$-\sin t = -1(\sin t) + 0(\cos t)$$

Thus,

$$\text{Second column} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

So, The Standard Matrix,

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\text{C.E.'s, } \lambda^2 - \text{Tr}(A)\lambda + |A| = 0$$

$$\therefore \lambda^2 - 0\lambda + 1 = 0$$

$$\therefore \lambda = i, -i$$

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for $\lambda = i$, $A - \lambda I = 1$

$$A - \lambda I = \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix}$$

$$R_2 \rightarrow iR_2 + R_1$$

$$\approx \begin{bmatrix} -i & -1 \\ 0 & 0 \end{bmatrix}$$

Rank = 1

$n - r = 1$ free variable

Let, $y = k$

So,

$$-ix - y = 0$$

$$\therefore x = -\frac{k}{i}$$

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$$\therefore x = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -k/i \\ k \end{bmatrix} = \frac{-k}{i} \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\therefore \text{functional basis} = \{ \sin t \ominus i \cos t \}$$

Similarly for $\lambda = -i$

$$\therefore \text{functional basis} = \{ \sin t \oplus i \cos t \}$$

$\rightarrow$ since $\lambda = i, -i$
so

Conjugate

(15) Let $T: S \rightarrow S$ is defined by $T(A) = BA + AB$, where $A \in S$, and $S = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} : a, b, c \in \mathbb{R} \right\}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Show that the distinct eigen values of T are $\lambda = 0, 2, -2$.

→ Given,

$$S = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix}, a, b, c \in \mathbb{R} \right\}$$

$$\text{basis are} = \left\{ \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{E_1}, \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{E_2}, \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{E_3} \right\}$$

Given,

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$$T(A) = BA + AB$$

$$T(E_1) = BE_1 + E_1B$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= E_2$$

$$= 0 \cdot E_1 + 1 \cdot E_2 + 0 \cdot E_3$$

$$= T(0, 1, 0) = M_1$$

Similarly,

for E_2 ,

$$T(E_2) = BE_2 + E_2B$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= 2E_1 + 2E_3$$

$$= T(2, 0, 2) = M_2$$

Similarly for E_3 ,

$$T(E_3) = B E_3 + E_3 B$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= E_2$$

$$= 0 \cdot E_1 + 1 \cdot E_2 + 0 \cdot E_3$$

$$= T(0, 1, 0) = M_3$$

now, The matrix is given by,

$$A = \begin{bmatrix} M_1 \\ M_2 \\ M_3 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

now, C.E. is,

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

where,

$$S_1 = 0$$

$$S_2 = -2 + 0 - 2 = -4$$

$$S_3 = 0$$

So,

$$\lambda^3 - 4\lambda = 0$$

$$\therefore \lambda(\lambda^2 - 4) = 0$$

$$\therefore \lambda = 0, 2, -2 \quad \underline{\text{Proved}}$$

for $\lambda = 0$, $AM = 1$

$$A - \lambda I = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$\approx \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G.M = n - r = 3 - 2 = 1 \text{ free variable}$$

$$\therefore A.M = G.M = 1 \text{ for } \lambda = 0$$

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Now Let, $z = K$

$$\text{So, } x + z = 0$$

$$\therefore x = -K$$

$$\therefore y = 0$$

Thus,

$$x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -K \\ 0 \\ K \end{bmatrix} = K \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore$ Eigen space for $\lambda = 0$

$$S_1 = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Similarly find for others. $\lambda = 2, -2$

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Minimal polynomial and Jordan Canonical Forms:

Monic Polynomial: A polynomial $f(t)$ is monic if its leading coefficient equals one.

Minimal polynomial: Let A be a square matrix. Let $J(A)$ denote the collection of polynomials for which A is a root that is, for which $f(A) = 0$. The set is not empty [because the Cayley-Hamilton theorem tells us that the characteristic polynomial belongs to $J(A)$]. Let $m(t)$ denote the monic polynomial of lowest degree in $J(A)$. We call $m(t)$ the minimal polynomial of the matrix.

- Suppose M is a block diagonal matrix with diagonal blocks $A_1, A_2, \dots, A_n$. Then the characteristic polynomial of M is the product of the characteristic polynomials of the diagonal blocks A_i .
- Suppose M is a block diagonal matrix with diagonal blocks $A_1, A_2, \dots, A_n$. Then the minimal polynomial of M is equal to the least common multiple of the minimal polynomials of the diagonal blocks A_i .

∴ find the minimal polynomial of.

$$(2.) \begin{bmatrix} 2 & 2 & -5 \\ 3 & 7 & -15 \\ 1 & 2 & -4 \end{bmatrix}$$

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∴ C.O.F is,

$$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0$$

where,

$$s_1 = 5$$

$$s_2 = 2 - 3 + 8 = 7$$

$$s_3 = 3$$

Thus,

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

$$\therefore \lambda = 3, 1, 1$$

So,

$$(\lambda - 3)(\lambda - 1)^2 = 0 \text{ is the characteristic equation.}$$

now, The characteristic polynomial is given by,

$$f(\lambda) = (\lambda - 3)(\lambda - 1)^2$$

now,

To find the minimal polynomial, firstly we have to find the possible divisions.

Let,

$$m_1(\lambda) = (\lambda - 3)(\lambda - 1)$$

$$m_2(\lambda) = (\lambda - 3)(\lambda - 1)^2$$

Consider, $m_1(x) = (x-3)(x-1)$

$$= (A - 3I)(A - I)$$

$$= \begin{bmatrix} -1 & 2 & -5 \\ 3 & 4 & -15 \\ 1 & 2 & -7 \end{bmatrix} \begin{bmatrix} 1 & 2 & -5 \\ 3 & 6 & -15 \\ 1 & 2 & -5 \end{bmatrix}$$

$$= 0$$

$\therefore m_1(x)$ is the minimal polynomial

b) $\begin{bmatrix} 4 & -2 & 2 \\ 6 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$

$\Rightarrow$ C.E. is.

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where,

$$S_1 = 4$$

$$S_2 = -1 + 6 + 0 = 5$$

$$S_3 = 2$$

$\Rightarrow$ Thus,

$$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$$

$$\therefore \lambda = 2, 1, 1$$

$\therefore (\lambda - 2)(\lambda - 1)^2 = 0$ is the characteristic eqn.

So,

characteristic polynomial,

$$f(\lambda) = (\lambda - 2)(\lambda - 1)^2$$

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$\Rightarrow$ Then,

possible divisor of $f(\lambda)$ are

$$m_1(x) = (x-2)(x-1)$$

$$m_2(x) = (x-2)(x-1)^2$$

Consider,

$$m_1(x) = (A - 2I)(A - I)$$

$$= \begin{bmatrix} 2 & -2 & 2 \\ 6 & -5 & 4 \\ 3 & -2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & -2 & 2 \\ 6 & -4 & 4 \\ 3 & -2 & 2 \end{bmatrix}$$

$$= 0 \Rightarrow m_1 \text{ is a minimal polynomial}$$

$$c) \begin{bmatrix} 3 & -2 & 2 \\ 4 & -4 & 6 \\ 2 & -3 & 5 \end{bmatrix}.$$

Char is,

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

where,

$$S_1 = 4$$

$$S_2 = -2 + 11 - 4 = 5$$

$$S_3 = 2$$

Thus,

$$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$$

$$\therefore \lambda = 2, 1, 1.$$

$$\therefore (\lambda - 2)(\lambda - 1)^2 = 0 \text{ is the characteristic eq.}$$

So, characteristic polynomial,

$$f(\lambda) = (\lambda - 2)(\lambda - 1)^2$$

Then,

The possible divisor of $f(\lambda)$ are

$$m_1(\lambda) = (\lambda - 2)(\lambda - 1)$$

$$m_2(\lambda) = (\lambda - 2)(\lambda - 1)^2$$

Consider,

$$M_1(\lambda) = (A - 2I)(A - I)$$

$$= \begin{bmatrix} 1 & -2 & 2 \\ 4 & -6 & 6 \\ 2 & -3 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 & -2 & 2 \\ 4 & -5 & 6 \\ 2 & -3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 2 & -2 \\ -4 & 4 & -4 \\ -2 & 2 & -2 \end{bmatrix} \neq 0$$

$\therefore M_1(\lambda)$ is not minimal polynomial

Thus, clearly $m_2(\lambda)$ is a minimal polynomial,

Since, By Cayley-Hamilton theorem, the sq. matrix satisfy its own characteristic eq.

d) $\begin{bmatrix} 3 & 1 & -1 \\ 2 & 4 & -2 \\ -1 & -1 & 3 \end{bmatrix}$.

→ C, E, a's

$$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - s_3 = 0$$

where,

$$s_1 = 10$$

$$s_2 = 10 + 8 + 10 = 28$$

$$s_3 = 24$$

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Thus,

$$\lambda^3 - 10\lambda^2 + 28\lambda - 24 = 0$$

$$\therefore \lambda = 6, 2, 2$$

$$\therefore (\lambda - 6)(\lambda - 2)^2 = 0 \text{ is a characteristic eq.}$$

So,

Characteristic polynomial,

$$f(\lambda) = (\lambda - 2)^2(\lambda - 6)$$

Then,

the possible divisors of $f(\lambda)$ are

$$m_1(\lambda) = (\lambda - 2)(\lambda - 6)$$

$$m_2(\lambda) = (\lambda - 2)^2(\lambda - 6)$$

Consider,

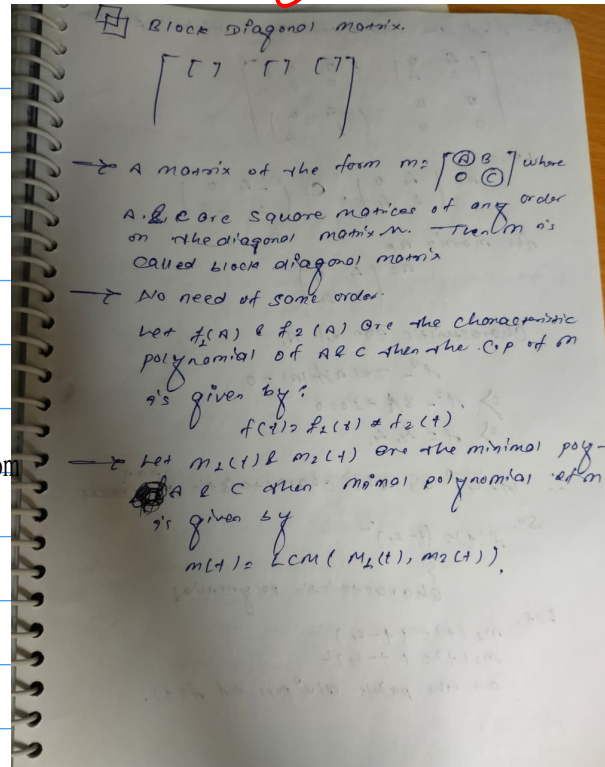
$$m_1(\lambda) = (A - 2I)(A - 6I)$$

$$= \begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ -1 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} -3 & 1 & -1 \\ 2 & -2 & -2 \\ -1 & -1 & -3 \end{bmatrix}$$

$$= 0$$

$\therefore m_1(\lambda)$ is characteristic polynomial.

Block Diagonal Matrix



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2. Find C.P & M.P

$$(i) \begin{bmatrix} \begin{pmatrix} 4 & 0 \\ 1 & 4 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} -4 & 0 \\ 1 & -4 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{bmatrix} = M$$

⇒ Here,

$$A = \begin{bmatrix} 4 & 0 \\ 1 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} -4 & 0 \\ 1 & -4 \end{bmatrix}$$

for matrix A ,

C.E is,

$$\lambda^2 - \text{Tr}(A)\lambda + |A| = 0$$

$$\text{or, } \lambda^2 - 8\lambda + 16 = 0$$

$$\therefore \lambda = 4, 4$$

$$\therefore (\lambda - 4)^2 = 0 \text{ is the characteristic eqn.}$$

So,

characteristic polynomial.

$$f(\lambda) = (\lambda - 4)^2$$

Then the possible divisors of $f(t)$,

$$m_1(t) = t - 4$$

$$m_2(t) = (t - 4)^2$$

Consider,

$$m_1(t) = (t - 4)$$

$$\Rightarrow m_1(A) = A - 4I$$

$$= \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \neq 0$$

$\therefore m_1(t)$ is not minimal polynomial of A

Hence,

$m_2(t)$ is a minimal polynomial of A .

for matrix C ,

$C \in \mathbb{R}$'s,

$$t^2 + 8t + 16 = 0$$

$$\therefore t = -4, -4$$

$\therefore (t + 4)^2 = 0$ is characteristic eqⁿ.

So,

characteristic polynomial,

$$f(t) = (t + 4)^2$$

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Then,

The possible divisors of $f(t)$

$$m_1(t) = (t + 4)$$

$$m_2(t) = (t + 4)^2$$

Consider,

$$m_1(t) = (t + 4)$$

$$\Rightarrow m_1(C) = C + 4I$$

$$= \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \neq 0$$

$\therefore m_1(t)$ is not minimal polynomial.

Hence,

$m_2(t)$ is a minimal polynomial.

Now, for matrix M .

Characteristic polynomial,

$$f(t) = (t-4)^2(t+4)^2 \\ = (t^2 - 16)^2$$

//

Similarly,

the minimal polynomial,

$$m(t) = \text{LCM}[(t-4)^2, (t+4)^2] \\ = (t-4)^2(t+4)^2 \\ = (t^2 - 16)^2$$

//

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$$\text{ii) } \begin{bmatrix} 3 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} = M$$

$$\Rightarrow A = \begin{bmatrix} 3 & 0 \\ 1 & 3 \end{bmatrix}, C = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

for matrix A ,

C.E.'s,

$$\lambda^2 - 6\lambda + 9 = 0$$

$$\therefore \lambda = 3, 3$$

$$\therefore (\lambda - 3)^2 = 0 \text{ is C.E.}$$

So,

Characteristic polynomial,

$$f(\lambda) = (\lambda - 3)^2$$

Then,

The possible divisor of $f(\lambda)$.

$$m_1(\lambda) = \lambda - 3$$

$$m_2(\lambda) = (\lambda - 3)^2$$

Consider,

$$m_1(\lambda) = \lambda - 3$$

$$\Rightarrow m_1(A) = A - 3I \\ = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \neq 0$$

$\therefore m_2(t)$ is not a minimal polynomial
Hence, $m_2(t)$ is a minimal polynomial.

For matrix C ,

$C \in \mathbb{R}$ is

$$\lambda^2 - 6\lambda + 9 = 0$$

$$\therefore \lambda = 3, 3$$

$$\therefore (\lambda - 3)^2 = 0 \text{ is } C \in \mathbb{R}.$$

So,

The character polynomial,
 $f(\lambda) = (\lambda - 3)^2$

Then,

possible divisor of $f(\lambda)$.

$$m_1(\lambda) = \lambda - 3$$

$$m_2(\lambda) = (\lambda - 3)^2$$

Consider,

$$m_1(\lambda) = \lambda - 3$$

$$\Rightarrow m_1(C) = C - 3I$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore m_1(\lambda)$ is a minimal polynomial

Thus,

characteristic polynomial of matrix M

$$\begin{aligned} f(\lambda) &= (\lambda - 3)^2 (\lambda - 3)^2 \\ &= (\lambda - 3)^4. \end{aligned}$$

Also,

minimal polynomial,

$$\begin{aligned} m(\lambda) &= \text{LCM}((\lambda - 3)^2, (\lambda - 3)^2) \\ &= (\lambda - 3)^2 // \end{aligned}$$

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$$\text{iii) } \begin{bmatrix} 2 & 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix} = M$$

Let,

$$A = \begin{bmatrix} 2 & 5 \\ 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 2 & 0 \\ 3 & 5 & 0 \\ 6 & 0 & 7 \end{bmatrix}$$

for matrix A,

$C \in \mathbb{R}^{n \times n}$,

$$\lambda^2 - 4\lambda + 4 = 0$$

$$\therefore \lambda = 2, 2$$

$\therefore (\lambda - 2)^2 = 0$ is the characteristic eqⁿ.

So,

$f(\lambda) = (\lambda - 2)^2$ is the char poly

Then,

The possible divisor of $f(\lambda)$

$$M_1(\lambda) = (\lambda - 2)$$

$$M_2(\lambda) = (\lambda - 2)^2$$

Consider, $M_1(\lambda) = \lambda - 2$

$$\Rightarrow M_1(A) = A - 2I$$

$$= \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix} \neq 0$$

$\therefore M_2(\lambda)$ is a minimal polynomial,

for matrix C.

$C \in \mathbb{R}^{n \times n}$,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where,

$$S_1 = 16$$

$$S_2 = 35 + 28 + 14 = 77$$

$$S_3 = 98$$

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Then,

$$\lambda^3 - 16\lambda^2 + 77\lambda - 98 = 0$$

$$\therefore \lambda = 2, 7, 7.$$

$$\therefore (\lambda - 2)(\lambda - 7)^2 = 0 \text{ is characteristic eqn}$$

So,

Characteristic polynomial.

$$f(\lambda) = (\lambda - 2)(\lambda - 7)^2$$

Then,

The possible divisor of $f(\lambda)$

$$m_1(\lambda) = (\lambda - 2)(\lambda - 7)$$

$$m_2(\lambda) = (\lambda - 2)(\lambda - 7)^2$$

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Consider

$$m_1(\lambda) = (\lambda - 2)(\lambda - 7)$$

$$\Rightarrow m_1(C) = (C - 2I)(C - 7I)$$

$$= \begin{bmatrix} 2 & 2 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \cdot \begin{bmatrix} -3 & 2 & 0 \\ 3 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= 0$$

$\therefore m_1(\lambda)$ is a minimal polynomial,

Thus,

Characteristic polynomial of matrix M

$$f(\lambda) = (\lambda - 2)^2 \cdot (\lambda - 2)(\lambda - 7)^2$$

$$= (\lambda - 2)^3 (\lambda - 7)^2$$

Similarly,

minimal polynomial.

$$m(\lambda) = \text{LCM}[(\lambda - 2)^2, (\lambda - 2)(\lambda - 7)^2]$$

$$= (\lambda - 2)^2 (\lambda - 7)$$

//

- 3) Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. Show that A and B have different characteristic polynomials (and so are not similar) but have the same minimal polynomial. Thus, nonsimilar matrices may have the same minimal polynomial.

→ for matrix A ,

C.E. is,

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

where,

$$S_1 = 4,$$

$$S_2 = 2 + 1 + 2 = 5$$

$$S_3 = 2$$

Thus,

$$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$$

$$\therefore \lambda = 2, 1, 1$$

$$\therefore (\lambda - 2)(\lambda - 1)^2 = 0 \rightarrow \text{C.E.}$$

So,

$$\text{C.P.} \rightarrow$$

$$f(\lambda) = (\lambda - 2)(\lambda - 1)^2$$

Then, possible divis

$$M_1(\lambda) = (\lambda - 2)(\lambda - 1)$$

$$M_2(\lambda) = (\lambda - 2)(\lambda - 1)^2$$

Consider,

$$M_1(\lambda) = (\lambda - 2)(\lambda - 1)$$

$$= (A - 2I)(A - I)$$

$$= \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= 0$$

$\therefore M_A(\lambda) = M_1(\lambda)$ is a minimal polynomial.

for matrix B ,

C.E. is,

$$S_1 = 5, S_2 = 2 + 2 + 4 = 8, S_3 = 4$$

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$$\text{Thus } \lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\therefore \lambda = 1, 2, 2$$

$$\therefore (\lambda - 1)(\lambda - 2)^2 = 0 \rightarrow C.E$$

So,

characteristic polynomial,

$$f(\lambda) = (\lambda - 1)(\lambda - 2)^2$$

Then,

The possible divisor of $f(\lambda)$

$$M_1(\lambda) = (\lambda - 1)(\lambda - 2)$$

$$M_2(\lambda) = (\lambda - 1)(\lambda - 2)^2$$

Consider,

$$M_1(\lambda) = (\lambda - 1)(\lambda - 2)$$

$$= (B - I)(B - 2I)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

$$= 0$$

$\therefore M_B(\lambda) = M_2(\lambda)$ is a minimal polynomial.

Clearly characteristic polynomial of both A & B are different but the minimal polynomial is same

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V. Jordan Canonical Forms:

V. Jordan Canonical Forms:

Let $T: V \rightarrow V$ be a linear operator whose characteristic and minimal polynomials are, respectively, $f(t) = (t - \lambda_1)^{n_1} \dots (t - \lambda_r)^{n_r}$ and $m(t) = (t - \lambda_1)^{m_1} \dots (t - \lambda_r)^{m_r}$.

where the λ_i are distinct scalars. Then T has a block diagonal matrix representation J in which each diagonal entry is a Jordan block $J_{ij} = J(\lambda_i)$. For each λ_i , the corresponding J_{ij} have the following properties:

- (i) There is at least one J_{ij} of order m_i ; all other J_{ij} are of order $\leq m_i$.
- (ii) The sum of the orders of the J_{ij} is n_i .
- (iii) The number of J_{ij} equals the geometric multiplicity of λ_i .
- (iv) The number of J_{ij} of each possible order is uniquely determined by T .

- 1) Find the Jordan canonical forms of T given by characteristic and minimal polynomial of T respectively $f(t) = (t-2)^4(t-5)^3$ and $m(t) = (t-2)^2(t-5)^3$.

Given, $f(t) = (t-2)^{\textcircled{4}}(t-5)^3$

$m(t) = (t-2)^{\textcircled{2}}(t-5)^3$

for $\lambda = 2$,

$A.M = 4$

$G.M = 2$

$\therefore$ possible partition of 4 with largest block 2: $2+2$

$\therefore$ Jordan blocks for $\lambda = 2$

$\Rightarrow J_2(2), J_2(2)$

$= \left\{ \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \right\}$

for $\lambda = 5$.

$A.M = 3$

$G.M = 3$

$\therefore$ Jordan block = $J_3(5)$ only

$= \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix}$

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$\therefore J = \text{diag}(J_2(2), J_2(2), J_3(5))$
 $\rightarrow$ diagonal matrix

- 2) Write Jordan canonical form or blocks if $f(t) = (t-2)^5$ is characteristic polynomial and $m(t) = (t-2)^2$ is the minimal polynomial.

Given,

$$f(t) = (t-2)^5$$

$$m(t) = (t-2)^2$$

only one eigen value.

$$\text{i.e. } \lambda = 2$$

$$A.M = 5$$

$$G.M = 2$$

Thus,

possible partition of 5 with largest block size 2

$$= 2 + 2 + 1$$

$$\therefore \text{Jordan blocks} = \text{diag} [J_2(2), J_2(2), J_1(2)] \\ = \left\{ \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, [2] \right\}$$

- 3) Write Jordan canonical form or blocks if $f(t) = (t-2)^2(t-3)$ is characteristic polynomial and $m(t) = (t-2)(t-3)$ is the minimal polynomial.

Given,

$$f(t) = (t-2)^2(t-3)$$

$$m(t) = (t-2)(t-3)$$

for $\lambda = 2$

$$A.M = 2$$

$$G.M = 1$$

$$\therefore \text{Jordan blocks} = \{ [2], [2] \}$$

for $\lambda = 3$

$$A.M = 1$$

$$G.M = 1$$

$$\therefore \text{Jordan blocks} = \{ [3] \}$$

$$\therefore J = \text{diag}(2, 2, 3) //$$

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- 4) Find the Jordan canonical forms of T given by characteristic and minimal polynomial of T respectively $f(t) = (t+1)^5(t-4)^2$ and $m(t) = (t+1)^3(t-4)^2$.

Given,

$$f(t) = (t+1)^5(t-4)^2$$

$$m(t) = (t+1)^3(t-4)^2$$

for $\lambda = -1$

$$A.M = 5$$

$$G.M = 3$$

possible partition $= 3 + 2$

$\therefore$ Jordan blocks $= J_3(-1), J_2(-1)$

$$= \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} //$$

for $\lambda = 4$,

$$A.M = 2$$

$$G.M = 2$$

$\therefore$ Jordan Block $= J_2(4)$

$$= \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$$

Hence,

$$J = \text{diag}(J_3(-1), J_2(-1), J_2(4))$$

- 5) Write Jordan canonical form or blocks if $D(t) = (t+3)^6$ is characteristic polynomial and $m(t) = (t+3)^2$ is the minimal polynomial.

Given,

$$D(t) = (t+3)^6$$

$$m(t) = (t+3)^2$$

$$\lambda = -3$$

$$A.M = 6$$

$$G.M = 2$$

$\therefore$ possible partition = $2+2+2$

$\therefore$ Jordan blocks = $J_2(-3), J_2(-3), J_2(-3)$

$$= \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix}, \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix}, \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix}$$

- 6) Write Jordan canonical form or blocks if $f(t) = (t-4)^4$ is characteristic polynomial and $m(t) = (t-4)^2$ is the minimal polynomial.

Given, $f(t) = (t-4)^4$

$$m(t) = (t-4)^2$$

$$\text{For } \lambda = 4$$

$$A.M = 4$$

$$G.M = 2$$

$\therefore$ possible partition = $2+2$

$\therefore$ Jordan blocks = $J_2(4), J_2(4)$

$$= \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}, \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$$

7) Find the Jordan canonical form of $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & 0 \\ 1 & -2 & 3 \end{bmatrix}$.

Given,

C.E. is,

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

where,

$$S_1 = 6$$

$$S_2 = 6 + 4 + 2 = 12$$

$$S_3 = 8$$

Thus,

$$\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$$

$$\therefore \lambda = 2, 2, 2$$

$$\therefore (\lambda - 2)^3 = 0$$

$$A.M = 3$$

for a.m

$$A - 2I = \begin{bmatrix} -1 & 2 & -1 \\ 0 & 0 & 0 \\ 1 & -2 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1$$

$$\approx \begin{bmatrix} -1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank} = 1$$

$$Q.M = 2$$

$$\therefore \text{possible partition} = 2 + 1$$

$$\therefore \text{Jordan blocks} = \text{diag}(J_2(2), 2)$$

$$= \begin{bmatrix} \boxed{2} & \boxed{1} & 0 \\ 0 & \boxed{2} & 0 \\ 0 & 0 & \boxed{2} \end{bmatrix} \text{ or } \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, [2]$$

8) Find the Jordan canonical form of

$$\begin{bmatrix} 2 & 1 & 0 & 0 \\ -1 & 4 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 4 \end{bmatrix}$$

→ Block diagonal with two identical blocks,

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$$

C.E.'s,

$$\lambda^2 - 6\lambda + 9 = 0$$

$$\therefore (\lambda - 3)^2 = 0$$

for $\lambda = 3$

$$A.M. = 2$$

for G.M.

$$A - \lambda I = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\text{Rank} = 1$$

$$G.M. = 1$$

$\therefore$ possible partition = $1 + 1$

$\therefore$ Jordan Block = $J_2(3), J_2(3)$

$$= \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}, \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$$

O.R.,

$$\begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Chill. 